MEC-109
Research Methods in Economics

Indira Gandhi
National Open University
School of Social Sciences

Block

1

RESEARCH METHODOLOGY: ISSUES AND PERSPECTIVES

Expert Committee

Course Coordinator: Prof. Narayan Prasad

Block Preparation Team

Units	Resource Person	IGNOU Faculty (Format, Language and Conent editing)
1	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi
2	Prof. S.G. Kulkarni Professor of Philosophy University of Hyderabad, Hyderabad	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi
3,4	Prof. D. Narshimha Reddy Rtd. Professor of Economics University of Hyderabad, Hyderabad	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi
5	Dr. Nausheen Nizami Assistant Professor in Economics Gargi College, University of Delhi New Delhi	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi

Block Editor (Content)

Prof. Prem Vashistha
NCAER, New Delhi

Print Production

Mr. Manjit Singh
Section Officer (Pub.)
SOSS, IGNOU, New Delhi

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MEC-109 RESEARCH METHODS IN ECONOMICS

After completion of Master's degree in Economics, many of you may intend to start your career as professional economist. As professional economists, you would be required to carry out the task of analyzing many specific economic situations and indicate their impact on economic policy framework. Some of you may also pursue research degree programmes. In order to perform these tasks, you need to be quipped with the various constituents of Research Methods and the different techniques applied in data collection/analysis. The present course aims to cater to this need. The theoretical perspectives that guides research, tools and techniques of data collection and methods of data analysis together constitute the research methodology. Quantitative Research and Qualitative Research are associated with different paradigms. Accordingly approaches to social enquiry varies. Mixed methods by combining of at least one quantitative method and one qualitative method in measurement, collection or analysis of data is increasingly being used to strengthen the validity of research findings. The present course deals with all these aspects. The course comprises of 6 Blocks.

Block 1: This block deals with the theoretical perspectives/foundations guiding both quantitative and qualitative research. The essence of three paradigms i.e. positivism and post positivism. Interpretitivism and critical theory associated with different research strategies and methodologies have been discussed. This block covers the entire breadth of main trends of development in the philosophy of science and debates in the methodology of economics. The block has been divided into 5 units. **Unit 1** deals with the conceptual foundation related to Research Methodology and its constituents, approaches to social enquiry, research strategy, research process, an elementary idea of hypothesis and measurement scales of variables. **Unit 2** is devoted to scientific methodology covering Positivists' verification approach, Pauper's critical rationalism, and Thomas Kuhan's paradigm's shift approach. The fundamental differences between Popper's and Positivists' views at the one hand, and differences between Popper's view and Kuhan's views on the other, have also been explained. Model of scientific explanation broadly within the framework of positivist theory of scientific method constitutes the core of **Unit 3.** Basic rules of formal reasoning, models of explanation, role of logical reasoning in the formulation of research problem, and the problems involved in systematic explanation of phenomenon have been discussed in this unit. **Unit 4** presents the view of different economists about the models of explanation in economics in the positivists mainstream framework and a brief discussion on the alternative methods of explanation. **Unit 5** throws light on essentials of interpretativism and critical theory paradigm guiding the frameworks of qualitative research. Based upon different ontological and epistemological positions, these paradigms analyse the nature of reality differently.

Block 2 on research design and measurement issues sets the tone and context to provide balanced treatment to quantitative and qualitative research. The block comprises of three units. Research design and mixed methods, the characteristics of quantitative methods and qualitative methods, sampling design, the various issues relating to measurement of variables have been covered in this block.

Block 3: In order to validate the economic theory, we need their empirical verification. Further, in order to examine disparity in income, the inequality measures are important. Similarly in order to describe the development status of an area in terms of several dimensions, we need to develop skill to construct composite index numbers. For all these purposes, good exposure to methods of regression models, in-equality measures and composite index is required. **Block 3** meets this requirement.

Block 4: In order to analyse the quantitative and qualitative data, more analytical techniques like Multi Variate Analysis: Factor Analysis, Canonical Correlation Analysis, Cluster Analysis, Correspondence Analysis, and Structural Equation Models are being increasingly used under either mono method or mix methods. **Block 4** explains all these techniques.

Block 5: A large range of data analysis techniques are used in carrying out qualitative research. However, three important methods namely participatory methods, content analysis, and action research – relevant for conducting the qualitative research studies in the area of Economics have been covered in this Block.

Block 6: The availability of data is crucial for undertaking research. For this, one is expected to be familiar with the different databases, and sources, the concepts used in data compilation and the agencies involved in data collection. **Block 6** focuses on these aspects of database of Indian Economy.

UNIT 1 RESEARCH METHODOLOGY: CONCEPTUAL FOUNDATION

Structure

1.0 OBJECTIVES

After going through this unit, you will be able to:

- Explain the research methodology and its constituents;
- State the various philosophical perspectives that guide research in social sciences;
- Describe various research approaches and strategies that are applied to answer the research questions;
- Describe the various steps involved in the research process; and
- Discuss the various types of research designs appropriate for different types of research.

1.1 INTRODUCTION

Research plays a significant role in human progress. It inculcates scientific and inductive thinking and promotes the development of logical thinking and organization. The role of research in several fields of applied economics has increased in modern times. As an aid to economic policy, it has gained importance for policy makers in understanding the complex nature of the functioning of the economy. It also occupies special significance in analysing problems of business and industry. Social scientists study the relationships among many variables seeking answers to complex issues through research. In short, research aids the process of knowledge formation and serves as an important source of providing policy suggestions to different business, government and social organization. The knowledge of the critical perspectives or philosophy of science, techniques of data collection and tools or methods of analyzing data are essential for undertaking research in a systematic manner. In this unit, we shall focus our attention on important constituents of research methodology i.e., research perspectives, research approaches and strategies and data types and its sources, hypothesis formulation and measurement scale of variables. At the initial stage of research, several questions may arise in your mind – what do we mean by research? How is the term 'research methodology' distinct from 'research techniques' or 'research methods'? Which type of research approach can be applied to a particular situation or context? What are the steps involved in the research process and how to design a research project? We shall take up these issues in this unit. Let us begin by explaining the term research methodology and its constituents.

1.2 RESEARCH METHODOLOGY AND ITS CONSTITUENTS

Research in common parlance refers to a search for knowledge. It can be defined as a scientific and systematic enquiry either to discover new facts or to verify old facts, their sequences, interrelationships, causal explanation and the adherence to natural laws governing them. It thus aims to discover the truth by applying scientific methods.

Research Methodology is a wider term. It consists of three important elements:

i) theoretical perspectives or orientation to guide research and logic of enquiry,

ii) tools and techniques of data collection, and

iii) methods of data analysis.

Research Methods, comprises of **research techniques** and **tools. Research techniques** refer to the practical aspects of collecting data and the way the information/data obtained/collected is organized and analysed. **Tools** are the instruments that are used for data collection and its analysis. It includes questionnaire/schedules, dairies, check lists, maps, photos, drawings etc. Census and survey methods are mainly used to collect quantitative data. In qualitative research, data is generated/complied by way of participant observation, semi structured interviews, life histories, experiments, pilot studies, scenarios etc. Data analysis involves a set of statistical techniques used in establishing relationships between the different variables and in evaluating the accuracy of the results.

Thus, methodology, methods and tools/techniques are three distinct elements of the research process. Any one of these three elements by itself may not be adequate in many situations. For instance, no data can be systematically collected without adequate knowledge of techniques of data collection. Similarly, data can not be explained without comprehending the philosophy or perspective behind the characteristics underlying the variables to which the data relates. A sound knowledge of statistical techniques is also necessary to analyse the data efficiently.

1.3 THEORETICAL PERSPECTIVES

Theoretical perspectives relate to theories of knowledge which lies within the domain of philosophy of social science. The key concept associated with the perspectives is the paradigm. Let us start to discuss with the concept of paradigm.

Paradigm: A paradigm is a comprehensive belief system, world view or framework that guides research and practice. It consists of

- At the basic or fundamental level, a philosophy of science that makes a number of assumptions about fundamental issues relating to nature and characteristics of truth or reality (ontology) and the theory of knowledge dealing with how can we know the things that exist (epistemology).
- World view, conceptual and theoretical framework that guides research and practices used in the field.
- General methodological prescriptions including tools to be used for information/data collection and data analysis to conduct the work within the paradigm.

The exact number of world views and the names associated with a particular paradigm vary from author to another but four paradigms in the context of research approaches in social sciences, are important:

- Positivism and post positivism
- Critical theory
- Interpretivism
- Pragmatism.

The peculiar features of these paradigms are:

- They differ on the question of reality,
- They offer different reasons or purposes for doing research,
- They use different types of tools for data collection and sources of information also vary,
- They resort different ways of deriving meaning from the collected data,
- They vary in the relationship between research and practice.

Some authors have classified these paradigms into three categories by re-naming them as:

Realism | Constructivism | Pragmatism

- Realism begins by assuming that there is a real world that is external to the experience of any particular person and the goal of research is to understand that world.
- Constructivism begins by assuming that everyone has unique experience and beliefs and it posits that no reality exists outside of those perceptions.
- Pragmatism considers realism and constructivism as two alternate ways to understand the world. However, the questions about the nature of reality are less important than questions about what is meant to act and experience the consequence of those actions.

The knowledge of all these perspectives enable a researcher to make a meaningful choice about

1) the research problem;
2) the research questions to investigate this problem;
3) the research strategies to answer these questions;
4) the approaches to social enquiry that accompany these strategies;
5) the concept and theory that direct the investigation;
6) the sources, forms and types of data;
7) the methods for collecting and analyzing the data to answer these questions.

The characteristics of positivism and post positivism as theoretical and methodological perspectives to scientific knowledge have been provided in the next unit i.e. Unit 2 of this course. Similarly the characteristics of interpretivisms and critical theory perspective to conduct social research will be taken up in Unit 5 of this course.

1.4 APPROACHES TO SOCIAL ENQUIRY

Broadly two types of approaches are used in conducting research in social sciences: quantitative and qualitative. The studies conducted within the perspective (framework) of positivism/post-positivism/realism generally resort the quantitative approach and are termed as 'quantitative research'. Quantitative research integrates purposes and procedures that are deductive, objective and generalized. Emphasis is laid on the construction of general theories which are applied universally. Well controlled procedures with large number of cases are followed in conducting the studies.

On other hand, the studies conducted within the perspective of critical theory and interpretivism paradigms are termed as qualitative research. By using induction as a research strategy, qualitative research creates the theory and discovery through flexible, emergent research designs. It tries to evolve meaning and interpretation based on closer contacts between researchers and the people they study. Thus qualitative research consists of purposes and procedures that integrate inductive, subjective and contextual approaches. Based on the above outlines on the types of research – we may say that there are two basic approaches to research viz., quantitative approach and qualitative approach.

Mixed methods research by combining quantitative methods and qualitative methods are being used in social sciences. Hence, mixed method design by integrating quantitative and qualitative approach has also emerged as an approach to social enquiry.

Quantitative approach can be further sub-classified into: inferential approach; experimental approach; and simulation approach.

In **Inferential approach**, database is established through survey method and inference is drawn about characteristics or relationship of variables. In **experimental approach**, greater control is exercised over research environment. Some variables are manipulated to observe their effect on other variables.

Simulation approach refers to the operation of a numerical model that represents the structure of a dynamic process. It involves the construction of an artificial environment in which relevant information/data can be generated. Given the values of initial conditions, parameters and exogenous variables, a simulation is run to represent the behaviour of the process over time.

Qualitative approach deals with the subjective assessment of attitudes, opinions and behaviour of respondents in the field. Results are generated either in non-quantitative form or in a form which are subjected to relatively less rigorous quantitative treatment. Various techniques like group discussions, projective techniques, in-depth interviews etc., are used. The typical characteristics of both these approaches may be summarized in the following Table:

Table 1.1: Typical Characteristics of Qualitative and Quantitative Approach

Characteristics	**Qualitative Approach**	**Quantitative Approach**
1) Typical data collection methods	Participant observation, semi-structured interviews, group discussion report cards etc.	Laboratory observations, questionnaire, schedule or structured interviews.
2) Formulation of questions and answers	Open/loosely specified questions and possible answers. Questions and answers are exchanged in two way communication between researcher and respondent.	Closed questions (hypothesis) and answer categories to be prepared in advance.
3) Selection of respondents	Information maximization guides the selection of respondent. Every respondent may be unique (key person).	Representativeness as proportion of population N. Randam sample selection, sample size according to assumptions about distribution in population N.
4) Timing of analysis	Parallel with data collection	After data collection
5) Application of standard methods of analysis	Descriptive methods of analysis are used. Mixed methods are also used.	Standard statistical methods are frequently used.
6) The role of theories in the analysis	Existing theories are typically used only as point of departure for the analysis. Theories are further developed by forming new concepts and relations. The contents of the new concepts are studied and illustrated. Practical application of theory is illustrated by cases.	A-priori deducted theories are opeationalised and tested on data. The process of analysis is basically deductive.

1.5 RESEARCH STRATEGIES

Four basic strategies can be adopted in social research depending upon the researcher's belief/reliance on perspective/paradigm about the nature of reality.

Paradigm	Research Strategy
Positivism	Induction
Post positivism/Realism	Deduction
Critical realism	Retroduction
Interpretativism	Abduction

Each strategy has different starting points (Lewis-Beck, 2004):

1) The inductive strategy begins with collection of data from which generalisation is made and can be used as an elementary explanation;

2) The deductive strategy starts out with a theory that provides a possible answer. The theory is tested in the context of a research problem by collection of relevant data;

3) The retroductive strategy starts out with a hypothetical model of a mechanism that could explain the occurance of a phenomenon under investigation;

4) The abductive strategy starts with laying the concepts and meanings that are contained in social quarters account of activities related to a research problem.

Check Your Progress 1

1) Distinguish between Research Methodology and Research Methods.

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2) How does the knowledge of research perspectives help a researcher to undertake research studies in social sciences?

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3) How is retroductive research strategy different from abductive strategy?

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